

Madhusudhan Reddy Pittu

Curriculum Vitae

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📁 mathrulestheworld.github.io

Summary

- Postdoctoral Associate at the Courant Institute (NYU) working broadly on various Algorithms and Optimization problems.
- Ph.D. in CS from CMU. Publications at top venues (STOC, FOCS, SODA, ICALP). Research spanning Clustering, Experimental Design/ Subset Selection, Constrained PCA / Dimensionality Reduction, and Optimization from Human Feedback.
- Bronze Medalist representing India at the 57th International Mathematical Olympiad (IMO).
- Hands-on LLM experience building end-to-end fine-tuning and evaluation pipelines, including learned prompt representations for improving reasoning.
- Studying unintended behavioral shifts caused by LLM fine-tuning, with implications for safer and more alignment-aware training setups.

Skills

Theory	Algorithm Design and Analysis, Combinatorial Optimization, Integer and Linear programming, Convex Optimization, Linear algebra, Probability and Statistics, ML theory.
ML / AI	PyTorch, HuggingFace Transformers, parameter-efficient fine-tuning (soft prompts, prefix tuning, LoRA), training loops, model evaluation, SLURM/HPC
Languages	Python, C++

Education and Experience

- 2025–2026 **Postdoctoral Associate**, *Courant Institute, New York University*, New York.
Host: Prof. Anupam Gupta
- 2021–2025 **Ph.D. in Computer Science**, *Carnegie Mellon University*, Pittsburgh.
Advisors: Prof. Anupam Gupta and Prof. David P. Woodruff
- 2017–2021 **B.Tech. in Computer Science and Engineering**, *Indian Institute of Technology Kharagpur*, India.
Advisor: Asst. Prof. Palash Dey

Recent Projects

- Spring 2026 **Prompt Boosting for LLM Reasoning.**
Developing methods to enhance Large Language Model reasoning capabilities using learned/ generated soft or hard token prompts. Implemented experimental pipelines, fine-tuning, and evaluated reasoning shifts on Qwen, Gemma models.
- Summer 2026 **Weird Generalization in LLM Fine-Tuning.**
Systematically studying unexpected downstream effects of fine-tuning Large Language Models. Identified empirical cases where targeted fine-tuning produces broader behavioral shifts, and developed hypotheses to explain when and why such changes occur, with implications for safer and more alignment-aware fine-tuning setups.

Industry Visits & Internships

- Spring 2025 **Student Researcher**, *Google*.
Hosts: Sreenivas Gollapudi, Kostas Kollias
- Summer 2023 **Non-Degree Visiting Student**, *Toyota Technological Institute at Chicago*.
Hosts: Ali Vakilian, Yury Makarychev, Julia Chuzhoy
- Fall 2022 **Visiting Graduate Student**, *Simons Institute for the Theory of Computing*.
Data-Driven Decision Processes program
- Summer 2020 **Visiting Student Research Internship (Remotely)**, *Georgia Tech*, Atlanta, USA.
Hosts: Prasad Tetali and Mohit Singh
- Summer 2019 **Narendra Internship**, *Indian Institute of Science*, Bengaluru.
Host: Arindam Khan

Selected Publications

Full publication list on page 3 and at mathrulestheworld.github.io.

- STOC '26 Combinatorial Optimization using Comparison Oracles
with Vincent Cohen-Addad, Tommaso d'Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
[Optimization from Human Feedback](#)
- ICALP '25 Guessing Efficiently for Constrained Subspace Approximation
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
[Constrained PCA / Dimensionality Reduction](#)
- SODA '24 Approximation Algorithms for Weighted Nash Social Welfare via Convex and Non-Convex Programs
with Adam Brown, Aditi Laddha, and Mohit Singh
[Fair Resource Allocation](#)
- FOCS '23 The Price of Explainability for Clustering
with Anupam Gupta, Ola Svensson, and Rachel Yuan
[Explainable Clustering](#)
- FOCS '22 Determinant Maximization via Matroid Intersection Algorithms
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
[Experimental Design / Subset Selection](#)
- SODA '22 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
[Computational Geometry](#)

Awards and Achievements

- Secured **Bronze Medal** representing India at the **57th International Mathematical Olympiad (IMO)**, 2016
- **IITKGP Foundation-USA International Internship Award**, 2020
- **Infosys Award** for excellent performance in International Olympiads, 2016
- **Best Solution Award** in IMO training camp, 2015

Complete Publication Record

Peer-Reviewed Journal Articles

- 2026 **A $(2+\varepsilon)$ -Approximation Algorithm for Maximum Independent Set of Rectangles.**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM Transactions on Algorithms, online publication, July 2, 2026.
- 2025 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs.**
with Adam Brown, Aditi Laddha, and Mohit Singh
Mathematical Programming, Series A, online-first, December 23, 2025. Journal version of the SODA 2024 paper.

Peer-Reviewed Conference Papers

- 2026 **Combinatorial Optimization using Comparison Oracles.**
with Vincent Cohen-Addad, Tommaso d’Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
58th ACM Symposium on Theory of Computing (**STOC**), pp. 1995–2006.
- 2025 **Max-Cut with Multiple Cardinality Constraints.**
with Yury Makarychev and Ali Vakilian
APPROX/RANDOM, LIPIcs 353, Article 13, pp. 13:1–13:21.
- 2025 **Guessing Efficiently for Constrained Subspace Approximation.**
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
52nd International Colloquium on Automata, Languages, and Programming (**ICALP**), LIPIcs 334, Article 29, pp. 29:1–29:20.
- 2024 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs.**
with Adam Brown, Aditi Laddha, and Mohit Singh
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 1307–1327.
- 2023 **The Price of Explainability for Clustering.**
with Anupam Gupta, Ola Svensson, and Rachel Yuan
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 1131–1148.
- 2022 **Determinant Maximization via Matroid Intersection Algorithms.**
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 255–266.
- 2022 **A 3-Approximation Algorithm for Maximum Independent Set of Rectangles.**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 894–905.
- 2020 **On Guillotine Separability of Squares and Rectangles.**
with Arindam Khan
APPROX/RANDOM, LIPIcs 176, Article 47, pp. 47:1–47:22.

Additional Public Preprints

- 2025 **An Algorithmic Upper Bound for Permanents via a Permanent Schur Inequality.**
with Aditi Laddha
Preprint, arXiv:2509.08121, 2025.
- 2022 **Efficient Determinant Maximization for All Matroids.**
with Adam Brown, Aditi Laddha, and Mohit Singh
Preprint, arXiv:2211.10507, 2022.

Selected Talks and Presentations

- 2026 Combinatorial Optimization using Comparison Oracles, STOC, June 24, 2026.
- 2026 Geometric Methods for Combinatorial Optimization Using Comparison Oracles, NYU Geometry Seminar, March 10, 2026 (video).
- 2025 Combinatorial Optimization using Comparison Oracles, Yale Theory Student Seminar, October 24, 2025.
- 2025 Max-Cut with Multiple Cardinality Constraints, APPROX, August 12, 2025.
- 2025 Guessing Efficiently for Constrained Subspace Approximation, ICALP, July 10, 2025.
- 2023 The Price of Explainability for Clustering, FOCS, November 7, 2023; also presented at CMU Theory Lunch, October 25, 2023.
- 2022 Efficient Determinant Maximization for All Matroids, IISc-MSR/Bangalore Theory Seminar, Indian Institute of Science, December 23, 2022 (video).
- 2022 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles, SODA 2022.
- 2020 On Guillotine Separability of Squares and Rectangles, APPROX/RANDOM, August 17, 2020 (video).